



Probability and Statistics

ES-111 Spring 2025

Assignment 5

Deadline of submission: 12.05.2023

Q1. Find the mean, median, and mode for the sample whose observations, 15, 7, 8, 95, 19, 12, 8, 22, and 14, represent the number of sick days claimed on 9 federal income tax returns. Which value appears to be the best measure of the center of these data? State reasons for your preference.

Q2. The tar contents of 8 brands of cigarettes selected at random from the latest list released by the Federal Trade Commission are as follows: 7.3, 8.6, 10.4, 16.1, 12.2, 15.1, 14.5, and 9.3 milligrams. Calculate (a) the mean; (b) the variance.

Q3. In the 2004-05 football season, University of Southern California had the following score differences for the 13 games it played. 11 49 32 3 6 38 38 30 8 40 31 5 36. Find (a) the mean score difference; (b) the median score difference.

Q4. If all possible samples of size 16 are drawn from a normal population with mean equal to 50 and standard deviation equal to 5, what is the probability that a sample mean $\bar{X}$ will fall in the interval from $\mu_{\bar{X}} - 1.9\sigma_{\bar{X}}$ to $\mu_{\bar{X}} - 0.4\sigma_{\bar{X}}$? Assume that the sample means can be measured to any degree of accuracy.

Q5. Given the discrete uniform population

$$f(x) = \begin{cases} \frac{1}{3}, & x = 2, 4, 6, \\ 0, & \text{elsewhere,} \end{cases}$$

find the probability that a random sample of size 54, selected with replacement, will yield a sample mean greater than 4.1 but less than 4.4. Assume the means are measured to the nearest tenth.

Q6. The heights of 1000 students are approximately normally distributed with a mean of 174.5 centimeters and a standard deviation of 6.9 centimeters. Suppose 200 random samples of size 25 are drawn from this population and the means recorded to the nearest tenth of a centimeter. Determine

(a) the mean and standard deviation of the sampling distribution of $\bar{X}$;

- (b) the number of sample means that fall between 172.5 and 175.8 centimeters inclusive;
- (c) the number of sample means falling below 172.0 centimeters.

Q7. If a certain machine makes electrical resistors having a mean resistance of 40 ohms and a standard deviation of 2 ohms, what is the probability that a random sample of 36 of these resistors will have a combined resistance of more than 1458 ohms?

Q8. The amount of time that a drive-through bank teller spends on a customer is a random variable with

a mean $\mu = 3.2$ minutes and a standard deviation $\sigma = 1.6$ minutes. If a random sample of 64 customers is observed, find the probability that their mean time at the teller's window is

- (a) at most 2.7 minutes;
- (b) more than 3.5 minutes;
- (c) at least 3.2 minutes but less than 3.4 minutes.

Q9 The mean score for freshmen on an aptitude test at a certain college is 540, with a standard deviation of 50. Assume the means to be measured to any degree of accuracy. What is the probability that two groups selected at random, consisting of 32 and 50 students, respectively, will differ in their mean scores by (a) more than 20 points? (b) an amount between 5 and 10 points?

Q10. A random sample of size 25 is taken from a normal population having a mean of 80 and a standard

deviation of 5. A second random sample of size 36 is taken from a different normal population having a mean of 75 and a standard deviation of 3. Find the probability that the sample mean computed from the

25 measurements will exceed the sample mean computed from the 36 measurements by at least 3.4 but

less than 5.9. Assume the difference of the means to be measured to the nearest tenth.

Q11. (a) Find $t_{0.025}$ when $\nu = 14$.

(b) Find $-t_{0.10}$ when $\nu = 10$.

(c) Find $t_{0.995}$ when $\nu = 7$.

Q12. (a) Find $P(T < 2.365)$ when $\nu = 7$.

(b) Find $P(T > 1.318)$ when $\nu = 24$.

(c) Find $P(-1.356 < T < 2.179)$ when $\nu = 12$.

(d) Find $P(T > -2.567)$ when $\nu = 17$.

Q13. Given a random sample of size 24 from a normal distribution, find k such that

(a) $P(-2.069 < T < k) = 0.965$;

(b) $P(k < T < 2.807) = 0.095$;

(c) $P(-k < T < k) = 0.90$.

Q14. A normal population with unknown variance has a mean of 20. Is one likely to obtain a random sample of size 9 from this population with a mean of 24 and a standard deviation of 4.1? If not, what conclusion would you draw?

Q15. A maker of a certain brand of low-fat cereal bars claims that the average saturated fat content is 0.5 gram. In a random sample of 8 cereal bars of this brand, the saturated fat content was 0.6, 0.7, 0.7, 0.3, 0.4, 0.5, 0.4, and 0.2. Would you agree with the claim? Assume a normal distribution.

Q16. Pull-strength tests on 10 soldered leads for a semiconductor device yield the following results, in

pounds of force required to rupture the bond:

19.8 12.7 13.2 16.9 10.6

18.8 11.1 14.3 17.0 12.5

Another set of 8 leads was tested after encapsulation to determine whether the pull strength had been increased by encapsulation of the device, with the following results:

24.9 22.8 23.6 22.1 20.4 21.6 21.8 22.5

Comment on the evidence available concerning equality of the two population variances.

Q17. A UCLA researcher claims that the life span of mice can be extended by as much as 25% when the calories in their diet are reduced by approximately 40% from the time they are weaned. The restricted diet is enriched to normal levels by vitamins and protein. Assuming that it is known from previous studies that $\sigma = 5.8$ months, how many mice should be included in our sample if we wish to be 99% confident that the mean life span of the sample will be within 2 months of the population mean for all mice subjected to this reduced diet?

Q18. Many cardiac patients wear an implanted pacemaker to control their heartbeat. A plastic connector module mounts on the top of the pacemaker. Assuming a standard deviation of 0.0015 inch and an approximately normal distribution, find a 95% confidence interval for the mean of the depths of all connector modules made by a certain manufacturing company. A random sample of 75 modules has an average depth of 0.310 inch.

Q19. A random sample of 100 automobile owners in the state of Virginia shows that an automobile is driven on average 23,500 kilometers per year with a standard deviation of 3900 kilometers. Assume the distribution of measurements to be approximately normal.

(a) Construct a 99% confidence interval for the average number of kilometers an automobile is driven annually in Virginia.

(b) What can we assert with 99% confidence about the possible size of our error if we estimate the average number of kilometers driven by car owners in Virginia to be 23,500 kilometers per year?

Q20. An efficiency expert wishes to determine the average time that it takes to drill three holes in a certain metal clamp. How large a sample will she need to be 95% confident that her sample

mean will be within 15 seconds of the true mean? Assume that it is known from previous studies that $\sigma = 40$ seconds.

Q21. A random sample of 12 graduates of a certain secretarial school typed an average of 79.3 words per minute with a standard deviation of 7.8 words per minute. Assuming a normal distribution for the number of words typed per minute, find a 95% confidence interval for the average number of words typed by all graduates of this school.

Q22. A random sample of 12 shearing pins is taken in a study of the Rockwell hardness of the pin head. Measurements on the Rockwell hardness are made for each of the 12, yielding an average value of 48.50 with a sample standard deviation of 1.5. Assuming the measurements to be normally distributed, construct a 90% confidence interval for the mean Rockwell hardness.

Q23. The following measurements were recorded for the drying time, in hours, of a certain brand of latex paint:

3.4 2.5 4.8 2.9 3.6

2.8 3.3 5.6 3.7 2.8

4.4 4.0 5.2 3.0 4.8

Assuming that the measurements represent a random sample from a normal population, find a 95% prediction interval for the drying time for the next trial of the paint.